

Romain Vo

📍 Paris ✉ voromain@gmail.com 🌐 website in romainvo 🌐 romainvo

Experience

Postdoctoral Researcher

CNRS, ENS de Lyon

Lyon, France

March 2025 - Sept 2026

I am developing learning-based methods for large-scale imaging inverse problems (e.g. 3D MRI, 3D CT, High-Resolution Radio Interferometry, Electron Microscopy, etc.) with a focus on memory-efficient algorithms and models.

- Part of a working group of the [NumPEX](#) project, focused on developing and benchmarking multi-gpus algorithms for large-scale imaging inverse problems.
- Maintainer of the [DeepInverse](#) library, an open-source project for solving imaging inverse problems with deep learning.

Ph.D. Student

CEA List & Mines Paris

Saclay, France

Oct 2021 - Jan 2025

In a department focused on Non-Destructive Testing (NDT) for industrial use-cases, my work was focused on building deep learning solutions for 3D Sparse-View X-ray Computed Tomography (CT).

- Building on existing variational optimization methods and advances in 3D graphics, I developed a novel memory-efficient approach for 3D X-Ray CT
- I proposed a convergent Plug-and-Play algorithm for 3D X-ray CT, which outperforms state-of-the-art methods in terms of reconstruction quality.
- Proposed a quantitative benchmark and analysis, under a common implementation, of advanced methods for sparse-view and low-dose X-ray CT.
- Trained deep learning models on HPC resources to solve inverse problems.
- Supervised 2 M.Sc. students on task-oriented reconstruction and the development of a new metric for evaluating the quality of 3D CT reconstructions.
- Peer reviewed 3 papers for a top-tier conference in computer vision (ECCV).

Research Engineer - Computer Vision and Deep Learning

IDEMIA

Osny, France

May 2021 - Aug 2021

I worked on biometric fraud detection (antispoofing) using deep learning methods.

- Setup an automatic data creation and acquisition pipeline to create new datasets for training and testing.
- Trained and developed deep learning solutions with interactive demos to showcase the models capabilities.

Research Intern - Computer Vision and Deep Learning

CEA List

Saclay, France

May 2020 - Nov 2020

I developed a deep learning algorithm for action detection in egocentric videos.

- Built and interfaced object detection, action recognition and video analysis models to build an end-to-end trainable model.
- Trained the models with HPC cluster on distributed GPUs across multiples nodes (SLURM/PyTorch).

Data Scientist Intern

Lely Holding N.V

Rotterdam, Netherlands

Jun 2019 - Sept 2019

Development of an animal health prediction model:

- Collect millions of data points from Azure Data Lake using SQL and Python
- Visualize and build insights from data to identify key features
- Develop a naive bayes classifier resulting in 25pnt % increase in recall and 5pnt % increase in precision.

Education

Mines Paris - PSL University

Ph.D. in Applied Mathematics

Paris, France

Oct 2021 - Jan 2025 (Expected)

My thesis focuses on building novel deep learning algorithms for sparse-view X-ray Computed Tomography (CT). Throughout the PhD, I have worked at the intersections of different fields such as inverse problem, optimization, computer vision, deep learning, variational inference and statistics, and have published in top-tier conferences.

IMT Atlantique (Institut Mines Télécom Atlantique)
Engineering Diploma - Computer Science for Decision Support

Courses in statistics, data analysis and visualization, optimization, mathematical planning and scheduling, machine learning and agile project management.

Nantes, France
Sept 2017 – Sept 2020

NTNU (Norwegian University of Science and Technology)
Erasmus Exchange

Trondheim, Norway
Jan 2019 – Jun 2019

Courses in machine learning, optimization and control, software architecture, and finance for engineering students.

Publications

Efficient Unrolled Networks for 3D Large-Scale Inverse Problems (Oral)	June 2026 <i>CVPR</i>
<u>Romain Vo</u> and Juliàn Tachella	  arXiv:2601.02141 
Learned Proximal Trajectory for 3D Sparse-View X-ray Computed Tomography	Sept 2024 <i>ECCV</i>
<u>Romain Vo</u> , Julie Escoda, Caroline Vienne, Étienne Decencière	 10.1007/978-3-031-72761-0_13 
Neural Field Regularization by Denoising for 3D Sparse-View X-ray Computed Tomography	Mars 2024 <i>3DV</i>
<u>Romain Vo</u> , Julie Escoda, Caroline Vienne, Étienne Decencière	 10.1109/3DV62453.2024.00094 
Evaluation and Comparison of Two Deep-Learning Strategies for On-Line X-Ray Computed Tomography	Jul 2021 <i>QNDE</i>
<u>Romain Vo</u> , Julie Escoda, Caroline Vienne, Étienne Decencière	10.1115/QNDE2022-98387 

Talks & Presentations

Memory-efficient reconstruction and task-based evaluation for sparse-view X-ray CT	June 2025 - Annecy, France <i>POPILSS 2025</i>
Learned Proximal Trajectory for 3D Sparse-View X-ray Computed Tomography	May 2024 - Fontainebleau, France <i>CMM - Mines paris</i>
Deep Learning and Sparse-View Computed Tomography for Material Science	Jan 2024 - Every, France <i>Center of Materials - Mines paris</i>
Neural Field Regularization by Denoising for Sparse-View X-ray Computed Tomography	Nov 2023 - Lyon, France <i>Workshop DIPOpt</i>
Neural Fields for Sparse-View X-ray Computed Tomography	Nov 2023 - Virtual, France <i>Ulm University Seminar</i>
Neural Fields for Sparse-View X-ray Computed Tomography	Jun 2023 - Toulon, France <i>ORASIS</i>
Deep Learning Approaches for Sparse-View X-ray Computed Tomography	May 2022 - Kaiserslautern, Germany <i>French-German Workshop</i>
Potentiel des Approches Deep Learning pour la Tomographie RX en ligne	May 2022 - Paris, Germany <i>Doctoriales COFREND</i>

Teaching

Deep Learning for Image Analysis	<i>Mines Paris</i>
Deep Learning for Image Analysis	<i>PSL University, IASD M.Sc</i>
Computer vision and ML for the material scientist (CVML)	<i>Mines Paris, Mastère DMS</i>
Medical Imaging – X-ray Computed Tomography	<i>Mines Paris, MAREVA</i>

Skills

My development setup: VSCode - Python - Jupyter notebooks - Git

Programming languages I have used: Java, SQL, C++, Matlab, HTML5/CSS3

Tools: Docker/Kubernetes, SLURM, Git, Jira, Obsidian

Languages: French (Native), English (Fluent)

References

Étienne Decencière - Research Director at Center for Morphological Mathematics, *Mines Paris*
etienne.decenciere@minesparis.psl.eu

Caroline Vienne - Deputy Head of the Interactive Robotics Department, *CEA List*
caroline.vienne@cea.fr